

Lab: Build to Think — Rapid Prototyping Your Invention

Objective

Experience making as cognition by building a rapid prototype of your invention idea using only readily available materials. You will practice fidelity matching, the perception-action loop, and bricolage, discovering how hands-on construction reveals insights that mental simulation alone cannot provide.

Materials / Prerequisites

- Your own invention idea or project
- Readily available materials: paper, cardboard, tape, scissors, pens/markers, rubber bands, paper clips, sticky notes, aluminum foil, string, or any other materials within reach
- A timer
- A smartphone camera (for documenting prototypes)
- Completed reading of Module 05: Creative Action

Important: This lab deliberately uses low-fidelity materials. The constraint is the point — work with what is at hand (bricolage). The most informative prototypes are often the cheapest and fastest to build. High-fidelity prototyping can actually be counterproductive in early stages because observers react to surface quality (finish, color, texture) rather than evaluating the core concept. By using paper, cardboard, and tape, you strip away the confounding variable of aesthetic finish and focus purely on function, form, and interaction — which are the critical uncertainties at this stage of invention.

Part 1: Define What You Are Testing (8 minutes)

Before building anything, identify the most critical uncertainty about your invention.

State your invention idea in one sentence:

Write your response here

List three major uncertainties about your invention:

Uncertainty	Type (Technical / User / Market / Aesthetic)	How Critical (1-10)

Write your response here

|

Write your response here

|

Write your response here

Write your response here

|

Write your response here

|

Write your response here

Write your response here

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Write your response here

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Write your response here

|

Select the most critical uncertainty. This is what your prototype will test:

Write your response here

What fidelity level is needed to test this uncertainty? (Circle one: Paper sketch / 3D foam model / Functional mechanism / Interactive facade / Other)

Write your response here

Part 2: First Build — Five-Minute Prototype (10 minutes)

This is where you experience making as cognition. The five-minute time constraint is not arbitrary — it is designed to prevent over-thinking and force you into the perception-action loop. Research on design cognition shows that inventors who start building within minutes of receiving a problem produce more creative solutions than those who plan extensively before building. The act of construction generates sensory evidence (how materials behave, how shapes fit together, how mechanisms feel in the hand) that mental simulation simply cannot provide.

Step 1 (5 min): Set a timer for exactly five minutes. Using only the materials at hand, build the simplest possible prototype that addresses your critical uncertainty. Do not plan — just build. The time constraint is intentional: it forces action over analysis.

Step 2 (5 min): Examine your prototype. Document it:

Describe what you built:

Write your response here

Take a photo of your prototype and note the filename:

Write your response here

Now answer these questions:

What did building reveal that thinking did not?

Write your response here

What prediction errors occurred during construction (things that did not work as you expected)?

What I Expected	What Actually Happened	What This Teaches Me

Write your response here

|

Write your response here

|

Write your response here

Write your response here

|

Write your response here

|

Write your response here

Write your response here

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Write your response here

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Write your response here

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Part 3: Iterate — Second Build (12 minutes)

The second prototype is where the real learning happens. Your first build was largely an act of externalization — getting ideas out of your head and into physical form. Your second build is an act of informed design — incorporating the prediction errors generated by the first build into a revised model. This is the perception-action loop completing its first full cycle: you acted (built), perceived (observed), updated your model (learned), and are now acting again (building version two). Each cycle should produce a measurably better prototype, not because you are refining aesthetics but because your generative model is becoming more accurate with each round of evidence.

Based on what you learned from the first prototype, build a revised version.

Step 1 (2 min): Identify the three most important changes to make:

1.

Write your response here

1.

Write your response here

1.

Write your response here

Step 2 (7 min): Build the second prototype, incorporating these changes. You may reuse materials from the first prototype (this is bricolage).

Step 3 (3 min): Compare the two prototypes:

Dimension	Prototype 1	Prototype 2	What Changed and Why
Core function			

Write your response here

|

Write your response here

|

Write your response here

| | **Size/shape** |

Write your response here

|

Write your response here

|

Write your response here

|| Usability |

Write your response here

|

Write your response here

|

Write your response here

|| What I learned |

Write your response here

|

Write your response here

|

Write your response here

|

Part 4: Use Simulation (15 minutes)

Building a prototype is only half the equation — using it is the other half. Use simulation is the practice of physically interacting with your prototype as if it were the final product, paying attention to the full experience: How do you pick it up? How do you hold it? What sequence of actions does it require? Where do your hands go? What can you see while using it? What feels awkward? Use simulation engages your motor and proprioceptive systems — cognitive resources that are unavailable during purely mental analysis — and generates embodied prediction errors that reveal design flaws invisible to the eye alone.

Step 1 (5 min): Take your second prototype and mime using it. Go through the complete usage scenario — pick it up, interact with it, complete the task it is designed for, put it away. Do this at least three times, each time paying attention to different aspects (ergonomics, sequence of operations, potential failure points).

What did you notice during use simulation that you did not notice during construction?

Write your response here

Step 2 (5 min): Show your prototype to at least one other person (in person or via photo/video). Ask them to mime using it or describe how they would use it. Watch what they do.

What did the other person do that you did not expect?

Write your response here

What prediction errors occurred between your model of usage and their actual behavior?

Write your response here

Step 3 (5 min): Based on the use simulation and observer feedback, identify the three most important insights:

1.

Write your response here

1.

Write your response here

1.

Write your response here

How would your third prototype differ from your second?

Write your response here

Part 5: Reflection — Making as Cognition (15 minutes)

This reflection phase is not optional — it is the "learning" phase of the perception-action-cognition-learning cycle that transforms raw experience into durable understanding. Without explicit reflection, the insights generated by building and testing will decay. Research on expert performance shows that deliberate reflection on performance — analyzing what worked, what did not, and why — is as important as the performance itself for skill development. Take this section seriously: the quality of your reflection determines how much your generative model actually updates from the prototyping experience.

Step 1 (5 min): Compare what you knew about your invention BEFORE this lab with what you know NOW. Fill in the table:

Aspect of My Invention	What I Believed Before Building	What I Know After Building	How Did Making Change My Understanding?
Core function			

Write your response here

|

Write your response here

|

Write your response here

| | **Form/shape** |

Write your response here

|

Write your response here

|

Write your response here

|| User interaction |

Write your response here

|

Write your response here

|

Write your response here

|| Key challenge |

Write your response here

|

Write your response here

|

Write your response here

|

Step 2 (5 min): Reflect on the role of materials in your creative process:

Did the available materials constrain your design? If so, how?

Write your response here

Did the materials suggest possibilities you had not considered? If so, what?

Write your response here

In what sense were the materials co-authors of your prototype?

Write your response here

Step 3 (5 min): Plan your next action step:

What is the most critical remaining uncertainty about your invention?

Write your response here

What prototype would you build next to address it, and at what fidelity level?

Write your response here

What materials or tools would you need?

Write your response here

Discussion / Debrief Questions

1. The module argues that "making is cognition, not just execution." Did this lab confirm or challenge that claim? What specific insight emerged from building that you are confident could not have emerged from thinking alone?
2. How did the five-minute time constraint in Part 2 affect your prototype? Did it produce a better or worse result than having unlimited time would have? What does this suggest about the role of time pressure in creative action?
3. Bricolage — building with whatever is at hand — forces the inventor to let materials guide the design. Did you experience this material-driven guidance? Give a specific example.
4. The perception-action loop cycled twice in this lab (first build, second build). How much did your generative model change between the first and second prototypes?

Would a third cycle have produced diminishing returns, or do you expect continued learning?

5. Jeff Hawkins carried a wooden block for months to simulate the Palm Pilot. What is the equivalent of a "wooden block prototype" for your invention — the simplest possible artifact that captures the core experience of your invention? Could you carry it with you?